

Parity-Resolved Deterministic-Numerator Coefficients

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Abstract

The exact RH-15 endpoint factorization separates the deterministic numerator into an even reduced Fredholm factor and an odd endpoint factor. We use this separation to give an all-order coefficient dictionary for the Hardy-scaled target. Every odd coefficient is an explicit scalar, while every even coefficient is a reduced-sector trace plus two explicit endpoint corrections. The formula reproduces the 27 RH-253 entries through order 28 with maximum floating residual below 6.5×10^{-14} . This is an anchor identity for the deterministic target, not a coefficient bridge for the moving cloud.

1 Factorization and notation

Let $r_H = 17/20$, $G_H(z) = G(z/r_H)$, and write

$$\log G_H(z) = - \sum_{n \geq 1} \frac{a_n}{n} z^n, \quad a_1 = 0.$$

RH-15 gives [2]

$$G(u) = e^{P_1 u} \det(I - u^2 T) A_*(u^2) B(u^2) C(u). \quad (1)$$

The logarithms are the branches normalized at zero. Put $b_n = \lambda^{-n}/(1 + \lambda^{-n})$ and $d_n = \lambda^{-2n}/(1 - \lambda^{-2n})$.

Theorem 1.1 (Parity-resolved anchor). *For every odd $n \geq 3$,*

$$a_n = (r_H \lambda)^{-n} (1 + \lambda^{-n})^{-1}. \quad (2)$$

For every $k \geq 1$,

$$a_{2k} = r_H^{-2k} \left[2 \operatorname{tr}(T^k) + \frac{2\lambda^{-2k}}{1 + \lambda^{-k}} - \frac{\lambda^{-2k}}{1 - \lambda^{-2k}} \right]. \quad (3)$$

Proof. The odd logarithm is $P_1 u + \log C(u)$. Since $P_1 = b_1$, its linear term cancels, and the coefficient of u^n for odd $n \geq 3$ is $-b_n/n$. Comparison with $-\sum a_n z^n/n$ after $u = z/r_H$ proves (2). For the even part, the coefficient of u^{2k} in the logarithm of the Fredholm factor is $-\operatorname{tr}(T^k)/k$. The deflated endpoint factor has coefficient $(b_k - \lambda^{-k})/k = -\lambda^{-2k}/(k(1 + \lambda^{-k}))$, and B contributes $d_k/(2k)$. Multiplication by $-2kr_H^{-2k}$ gives (3). $\square$

Corollary 1.2 (Odd tail identity). *For odd $n \geq 3$, the coefficient is positive and obeys*

$$0 < a_n < (r_H \lambda)^{-n}.$$

Thus odd target tails are controlled without a boundary supremum.

2 Finite cross-check and its scope

The experiment reads the archived RH-253 physical traces and evaluates (2)–(3); it does not fit a new rate. The 27 rows (13 odd and 14 even) give a maximum absolute residual below 6.5×10^{-14} in double precision. The RH-13 Arb certificate supplies the operator bounds needed for a separate all-order tail majorant [1].

The identity does not say that noisy cloud-selected traces converge to these numbers. It also does not construct a selector, a self-adjoint operator, or a zeta divisor. In particular, the finite cross-check is not an all-order cloud fit. Gates A–E remain false/open.

3 Route ledger

The target-tail and boundary-constant obligations are true after RH-262; the legal anchored head, cloud coefficient bridge, and uniform quotient tail are false/open. The next paper uses (3) together with the RH-13 trace-ideal bounds to certify a direct order-29 tail, retaining the parity split rather than applying a coarse global boundary constant.

References

- [1] Bin Wang. Validated reduced-sector spectral gaps. *Preprint*, 2026. RH-13 preprint.
- [2] Bin Wang. Parity-extracted bulk scattering. *Preprint*, 2026. RH-15 preprint.